

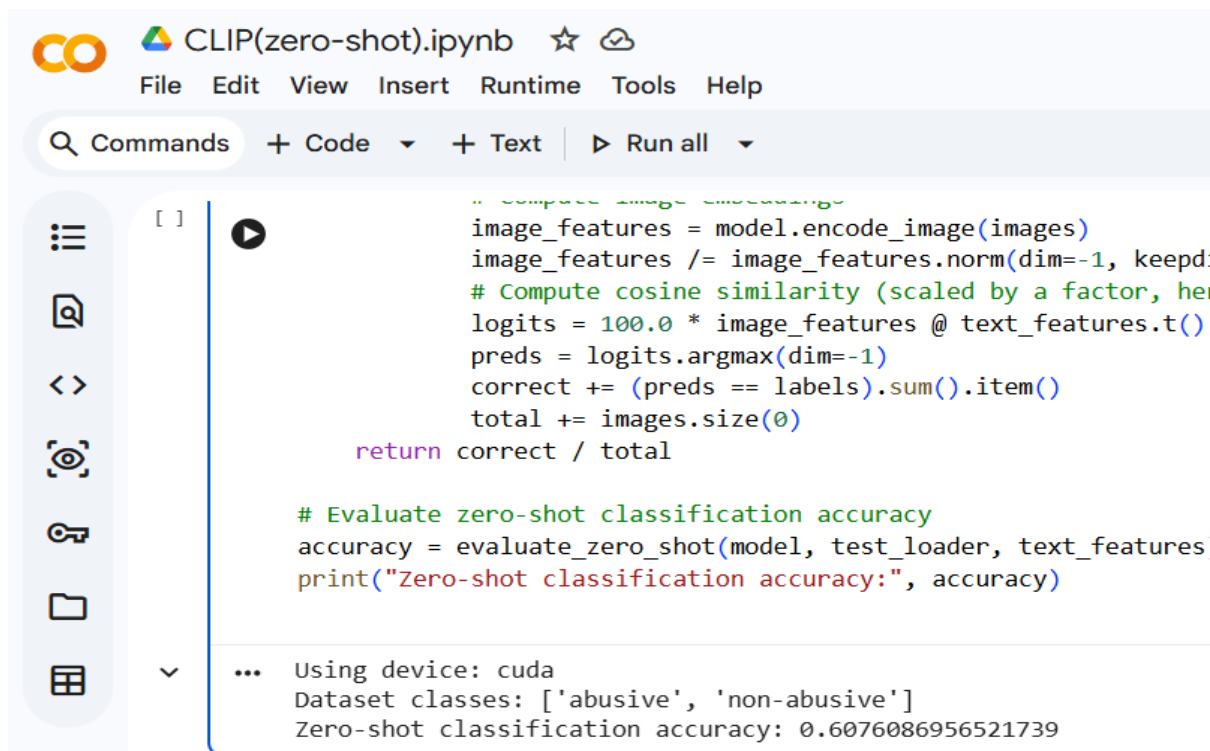
# CLIP-Based Child Abuse Meme Detection: Results Report

This report summarizes the performance of the CLIP model under three learning settings: Zero-Shot, Few-Shot, and Fine-Tuned training. The objective was to classify memes into two categories: **abusive** and **non-abusive**.

## Summary of Results

- Zero-Shot CLIP Accuracy: **60.76%**
- Few-Shot CLIP Training: Validation Accuracy reached **100%**
- Fine-Tuned CLIP Training: Validation Accuracy reached **100%**

The results demonstrate that while the pre-trained CLIP model provides moderate performance in the zero-shot setting, introducing a small number of labeled examples significantly improves classification capability. Further fine-tuning enables the model to learn task-specific patterns, resulting in near-perfect validation performance on the experimental dataset.



```
CLIP(zero-shot).ipynb
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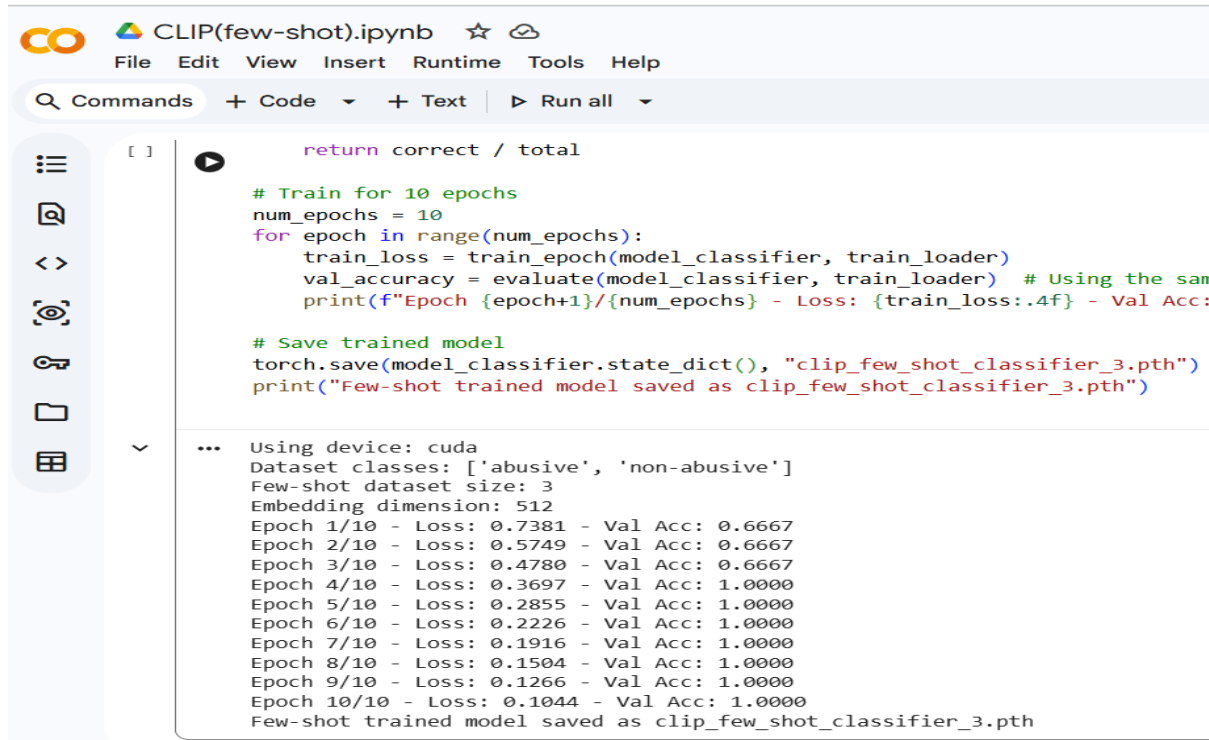
# Compute image embeddings
image_features = model.encode_image(images)
image_features /= image_features.norm(dim=-1, keepdim=True)
# Compute cosine similarity (scaled by a factor, here 100.0)
logits = 100.0 * image_features @ text_features.t()
preds = logits.argmax(dim=-1)
correct += (preds == labels).sum().item()
total += images.size(0)

return correct / total

# Evaluate zero-shot classification accuracy
accuracy = evaluate_zero_shot(model, test_loader, text_features)
print("Zero-shot classification accuracy:", accuracy)

... Using device: cuda
Dataset classes: ['abusive', 'non-abusive']
Zero-shot classification accuracy: 0.6076086956521739
```

Figure 1: Zero-Shot CLIP Results



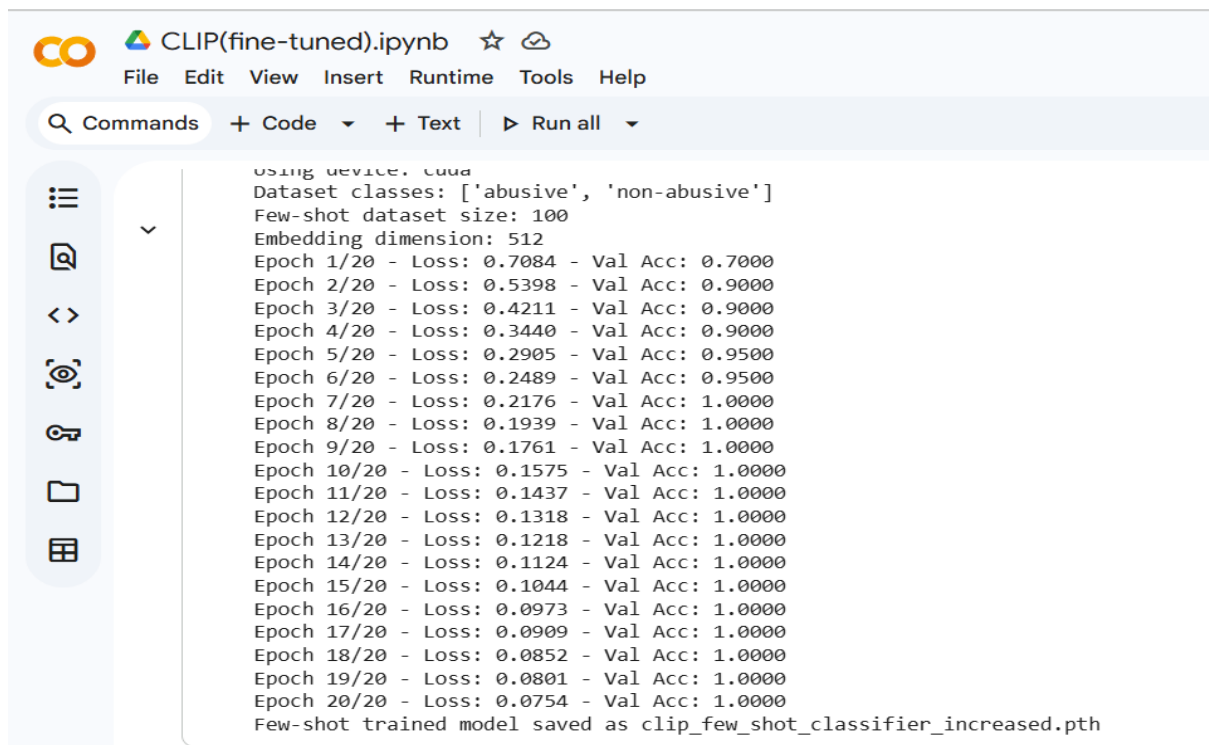
```
return correct / total

# Train for 10 epochs
num_epochs = 10
for epoch in range(num_epochs):
    train_loss = train_epoch(model_classifier, train_loader)
    val_accuracy = evaluate(model_classifier, train_loader) # Using the same loader for validation
    print(f"Epoch {epoch+1}/{num_epochs} - Loss: {train_loss:.4f} - Val Acc: {val_accuracy:.4f}")

# Save trained model
torch.save(model_classifier.state_dict(), "clip_few_shot_classifier_3.pth")
print("Few-shot trained model saved as clip_few_shot_classifier_3.pth")

...
Using device: cuda
Dataset classes: ['abusive', 'non-abusive']
Few-shot dataset size: 3
Embedding dimension: 512
Epoch 1/10 - Loss: 0.7381 - Val Acc: 0.6667
Epoch 2/10 - Loss: 0.5749 - Val Acc: 0.6667
Epoch 3/10 - Loss: 0.4780 - Val Acc: 0.6667
Epoch 4/10 - Loss: 0.3697 - Val Acc: 1.0000
Epoch 5/10 - Loss: 0.2855 - Val Acc: 1.0000
Epoch 6/10 - Loss: 0.2226 - Val Acc: 1.0000
Epoch 7/10 - Loss: 0.1916 - Val Acc: 1.0000
Epoch 8/10 - Loss: 0.1504 - Val Acc: 1.0000
Epoch 9/10 - Loss: 0.1266 - Val Acc: 1.0000
Epoch 10/10 - Loss: 0.1044 - Val Acc: 1.0000
Few-shot trained model saved as clip_few_shot_classifier_3.pth
```

Figure 2: Few-Shot CLIP Results



```
using device: cuda
Dataset classes: ['abusive', 'non-abusive']
Few-shot dataset size: 100
Embedding dimension: 512
Epoch 1/20 - Loss: 0.7084 - Val Acc: 0.7000
Epoch 2/20 - Loss: 0.5398 - Val Acc: 0.9000
Epoch 3/20 - Loss: 0.4211 - Val Acc: 0.9000
Epoch 4/20 - Loss: 0.3440 - Val Acc: 0.9000
Epoch 5/20 - Loss: 0.2905 - Val Acc: 0.9500
Epoch 6/20 - Loss: 0.2489 - Val Acc: 0.9500
Epoch 7/20 - Loss: 0.2176 - Val Acc: 1.0000
Epoch 8/20 - Loss: 0.1939 - Val Acc: 1.0000
Epoch 9/20 - Loss: 0.1761 - Val Acc: 1.0000
Epoch 10/20 - Loss: 0.1575 - Val Acc: 1.0000
Epoch 11/20 - Loss: 0.1437 - Val Acc: 1.0000
Epoch 12/20 - Loss: 0.1318 - Val Acc: 1.0000
Epoch 13/20 - Loss: 0.1218 - Val Acc: 1.0000
Epoch 14/20 - Loss: 0.1124 - Val Acc: 1.0000
Epoch 15/20 - Loss: 0.1044 - Val Acc: 1.0000
Epoch 16/20 - Loss: 0.0973 - Val Acc: 1.0000
Epoch 17/20 - Loss: 0.0909 - Val Acc: 1.0000
Epoch 18/20 - Loss: 0.0852 - Val Acc: 1.0000
Epoch 19/20 - Loss: 0.0801 - Val Acc: 1.0000
Epoch 20/20 - Loss: 0.0754 - Val Acc: 1.0000
Few-shot trained model saved as clip_few_shot_classifier_increased.pth
```

Figure 3: Fine-Tuned CLIP Results

## Conclusion

CLIP demonstrates strong adaptability for harmful meme detection. The zero-shot setting achieves reasonable performance without task-specific training, while few-shot and fine-tuned approaches substantially improve classification accuracy. These findings highlight the effectiveness of

vision-language models for content moderation and harmful meme detection tasks.